

Game Design Portfolio - Part 2

Signal Aftershock: Triangulation Race

A thematic game redesign based on Week 3: Triangle Territory

Format	2-3 player hotseat browser game prototype
Playable Prototype	https://bobmai624.github.io/signal-aftershock-triangulation-race-site/?v=e571d4b
Student	Bowen Mai
Subject	Game Design Portfolio - Part 2

Thematic Game Design

Executive Summary and Design Origin

Signal Aftershock: Triangulation Race develops my Week 3 workshop activity, Triangle Territory. The earlier design explored how dice-based uncertainty could create strategy in a line-drawing triangle-capture game. My original insight was that uncertainty should not replace decision-making; it should create conditions that make each decision harder and more meaningful. This redesign keeps that insight but turns the abstract territory contest into a disaster-response game about restoring emergency communications after an earthquake.

WEEK 3 WORKSHOP ACTIVITY — UNCERTAIN OUTCOMES

Triangle Territory

A dice-constrained territory game where players connect grid points to claim triangles and limit an opponent's options.

This design explores how uncertainty can create meaningful strategy rather than replace it.

Core Design Insight

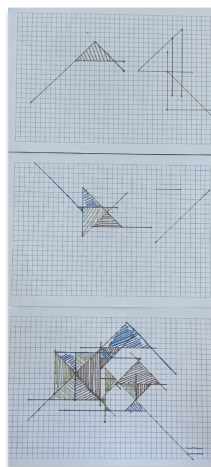
The game shows that uncertainty does not remove strategy; it creates the conditions for meaningful decisions. Dice rolls constrain line length, but players still choose whether to score, block, or build future territory.

How the Game Produces Uncertain Outcomes

Chance — A die roll determines the line length available each turn, preventing players from fully planning every move.
Decision-making — Players choose whether to score immediately, set up future triangles, or block an opponent's space.
Positional uncertainty — Every new line changes the value of nearby points and edges, making future opportunities hard to predict.

Reflection: Uncertainty as Design Tool

Zombie Dice helped me understand that chance becomes interesting when players still control how they respond to risk. In Triangle Territory, the die creates a constraint, while spatial choices create strategy. Testing showed that simple rules can produce emergent gameplay when every action reshapes the board.



Territory development test

MGE Takeaway

Mechanics — Dice, grid points, line drawing, and triangle capture.

Gameplay — Players adapt, score, block, and compete over shrinking space.

Experience — Chance creates tension, but decisions still shape the outcome.

Design Development

Initial concept — A grid-based area control game built around drawing lines to form triangles.

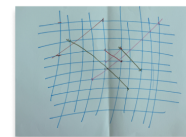
Testing — Different grid sizes, die values, and line-length limits were tested for fairness and strategy.

Refinement — Non-overlap and protected territory rules made claimed areas clearer and conflict more meaningful.

Final direction — A dice-constrained territory game where uncertainty shapes, but does not replace, player decision-making.



Initial workshop sketch



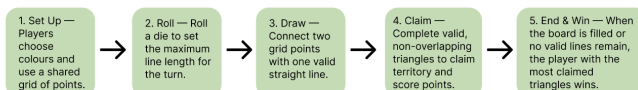
Early grid test



Triangle Capture Mechanic

A player claims territory when a newly drawn line closes a valid triangle between grid points. Each claimed triangle scores 1 point and cannot be overlapped, covered, or claimed again.

How to Play



Note: The die does not move a player token; it creates a drawing constraint.

Week 3 workshop source image: the original Triangle Territory portfolio activity established dice-constrained line drawing, triangle capture, and the design insight that uncertainty can create strategy rather than replace it.

The thinking process began by identifying which parts of the Week 3 activity were strongest. The die did not move a token; it created a drawing constraint. The grid points created a shared space of possible connections. The triangle closure rule turned a simple line into a scoring moment. I kept these three elements, then asked what theme could make them feel motivated rather than purely geometric. Emergency signal recovery fit the structure: an unstable die result becomes limited repair capacity, a drawn line becomes a temporary relay, and a completed triangle becomes a stable coverage zone.

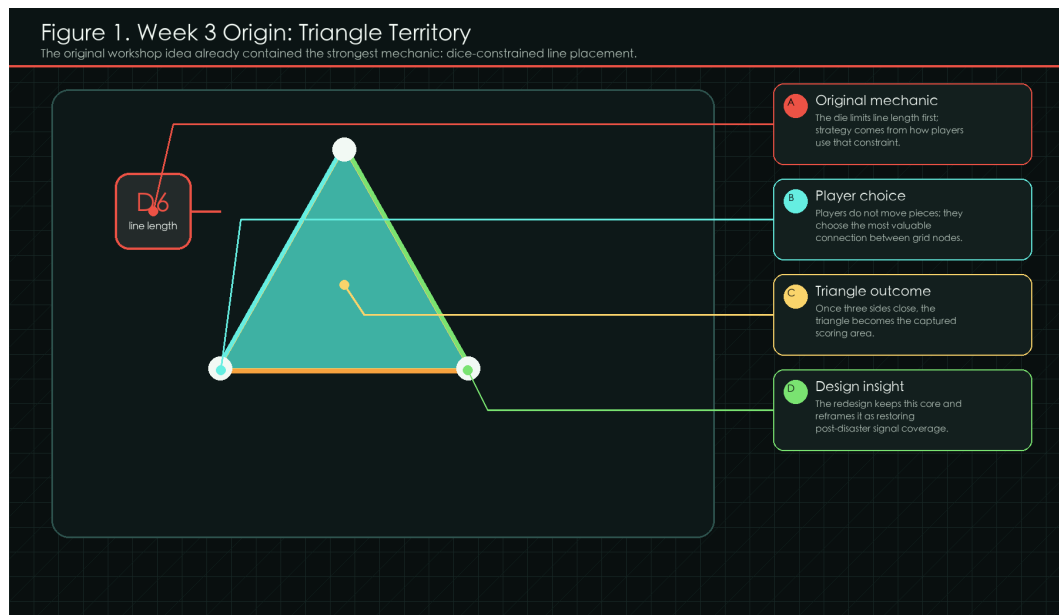


Figure 1. Week 3 origin mechanic: the annotated diagram shows how the original uncertainty mechanic becomes the foundation of the redesign.

The new game is a two- or three-player hotseat browser prototype. After the 04:17 aftershock, hospitals, shelters, relay towers, power stations, and supply points are losing contact. Different emergency signal teams must rebuild temporary relay lines, enclose stable coverage zones, and reconnect critical sites before time runs out. The player experience is a live emergency command map: quick turns, unstable resources, glowing routes, visible coverage zones, and urgent infrastructure icons. The winner has the highest Coverage Score, calculated from claimed area, critical-site bonuses, and full-length closure bonuses.

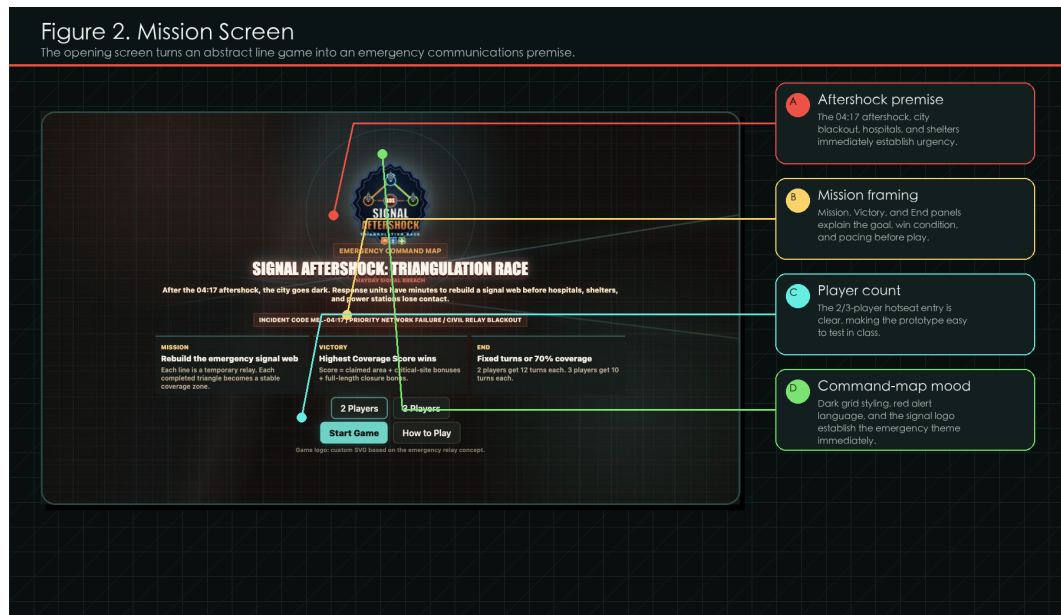


Figure 2. Mission screen: the opening interface frames the abstract line game as an emergency communications premise before play begins.

Core Loop and Player Interaction

The core loop is roll, draw, resolve, score, and pass. At the start of a turn, the active player rolls a die, which gives the available signal length. The player connects two grid points with a legal straight line on the triangular lattice. A legal line must follow one of the grid directions and cannot overlap an existing unit edge. Once placed, the line becomes a permanent relay, so every move changes both the current scoring opportunity and the future geometry of the city network.

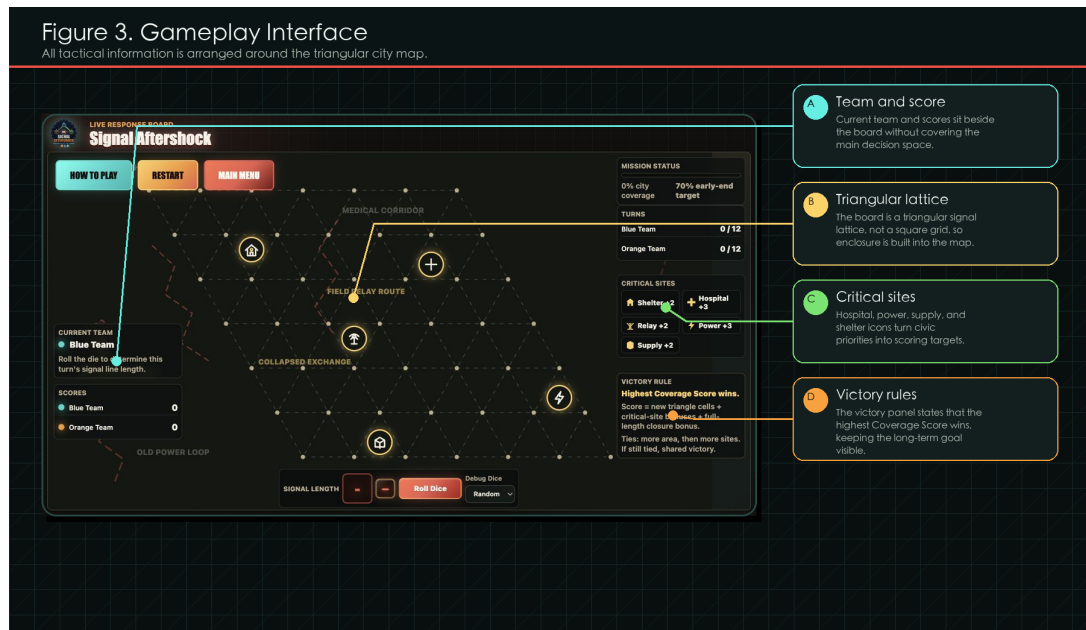


Figure 3. Gameplay interface: all tactical information is arranged around the triangular city map, including turn state, score, critical sites, and victory rules.

When a new line completes a triangle, the enclosed area becomes stable communication coverage. Scoring combines claimed triangle cells with bonuses for enclosed critical sites: Shelter +2, Relay +2, Supply +2, Hospital +3, and Power +3. A closure that uses the full rolled length gains an additional reward, encouraging players to balance safety against ambition. The game ends after fixed turns or when 70% of the map has been covered. Two-player games give each player 12 turns, and three-player games give each player 10 turns.

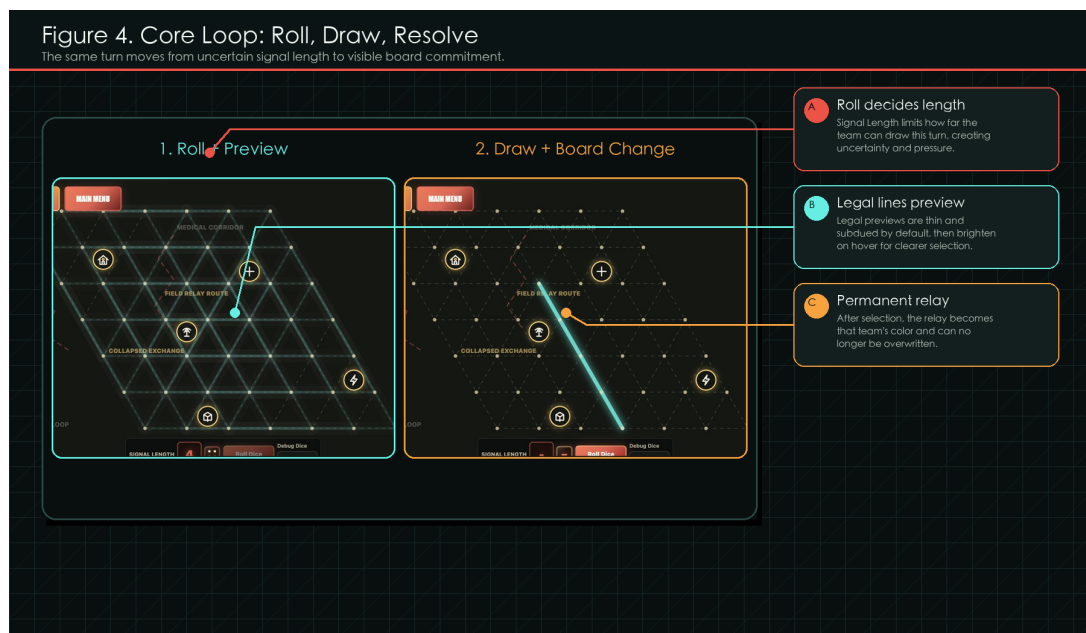


Figure 4. Core loop: a turn moves from uncertain signal length to player placement and then to a visible board commitment.

Player interaction is indirect competition through shared space. Players do not attack with cards or character powers. Instead, they block routes, compete for perimeter majority, and time closures. Ownership is decided by perimeter majority, not simply by the player who draws the final line. If Blue controls two sides of a triangle and Orange draws the third side, Blue still claims the zone; the closing player only wins when boundary control is tied. This prevents cheap last-move stealing and rewards long-term positioning.

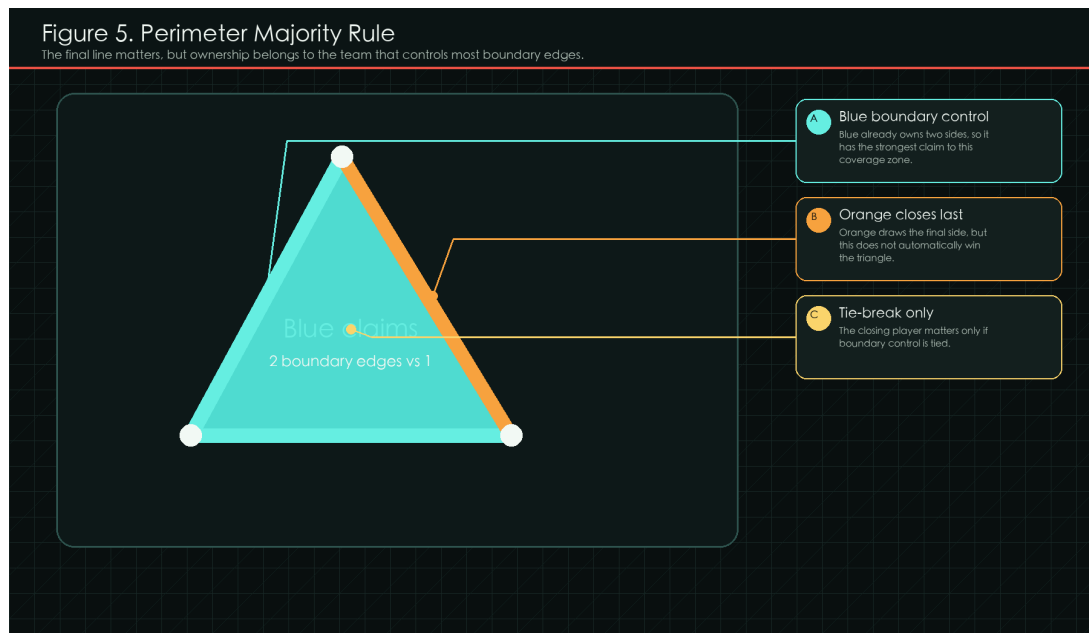


Figure 5. Perimeter majority rule: the final line matters, but ownership belongs to the team that controls most boundary edges.

Mechanics Design and Meaningful Play

The design builds on Dots and Boxes, where completed boundaries create ownership, but replaces square boxes with variable-size triangles (Barile n.d.). It also draws on Qix's attraction of claiming larger territory and Sim's demonstration that simple lines and triangles can generate strategic depth (Taito 1981; Simmons 1969).

The main mechanic is dice-controlled signal length. In Week 3, dice created uncertain drawing opportunities. In the redesign, that uncertainty becomes thematic: after a disaster, teams cannot fully control cable access, drone range, or repair capacity. However, the die never removes agency. A low roll can make a precise closure; a high roll can cross the city, build a large perimeter, or accidentally give another team an opportunity. This creates meaningful play because each action has an immediate, readable, and systemically integrated result (Salen and Zimmerman 2003).

Triangle size also matters. The system recognises large triangular enclosures, not only the smallest cells. Larger triangles claim more city area, so a player who plans a wider signal boundary is rewarded for the full scale of that achievement. If multiple closures are created at once, the largest valuable completed area is prioritised. This makes the scoring logic match the fiction: restoring a larger district should matter more than stabilising a tiny pocket.

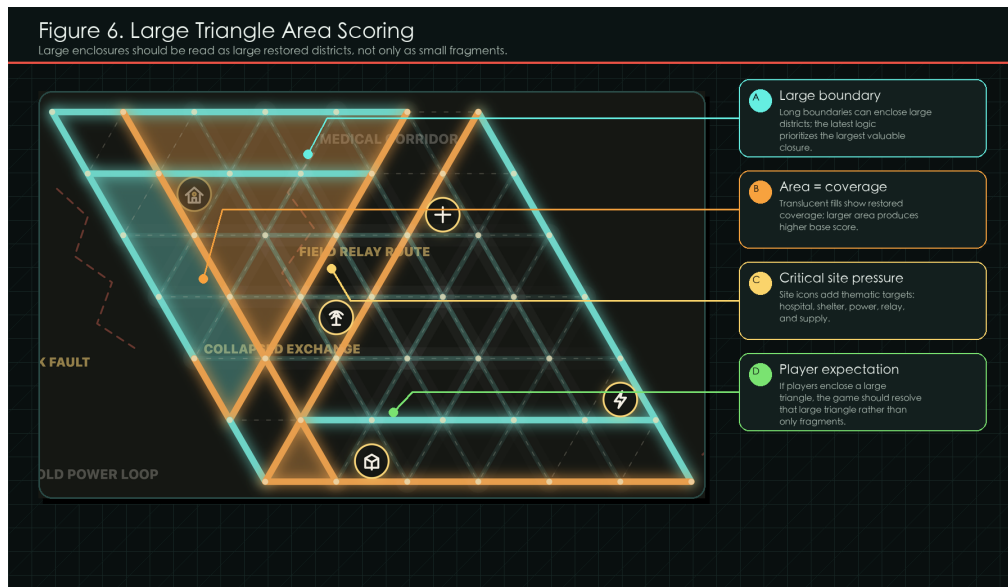


Figure 6. Large triangle area scoring: large enclosures are read as large restored districts, not only as small fragments.

Theme, Premise, and Alignment

The theme is not decoration layered on top of geometry; it explains the geometry. A line is a temporary relay. A completed triangle is a stable coverage zone because triangular redundancy suggests a stronger network than a single route. Claimed area represents restored communication coverage. Critical sites create narrative pressure because hospitals, shelters, power, relay, and supply targets are more urgent than empty space.

This alignment changes how players read their choices. They are not only asking, "How do I get points?" They are also asking whether they should reconnect the hospital now, secure a smaller shelter, or risk building toward a larger district. The theme gives emotional meaning to abstract spatial decisions, while the mechanics make the emergency premise playable rather than only descriptive.

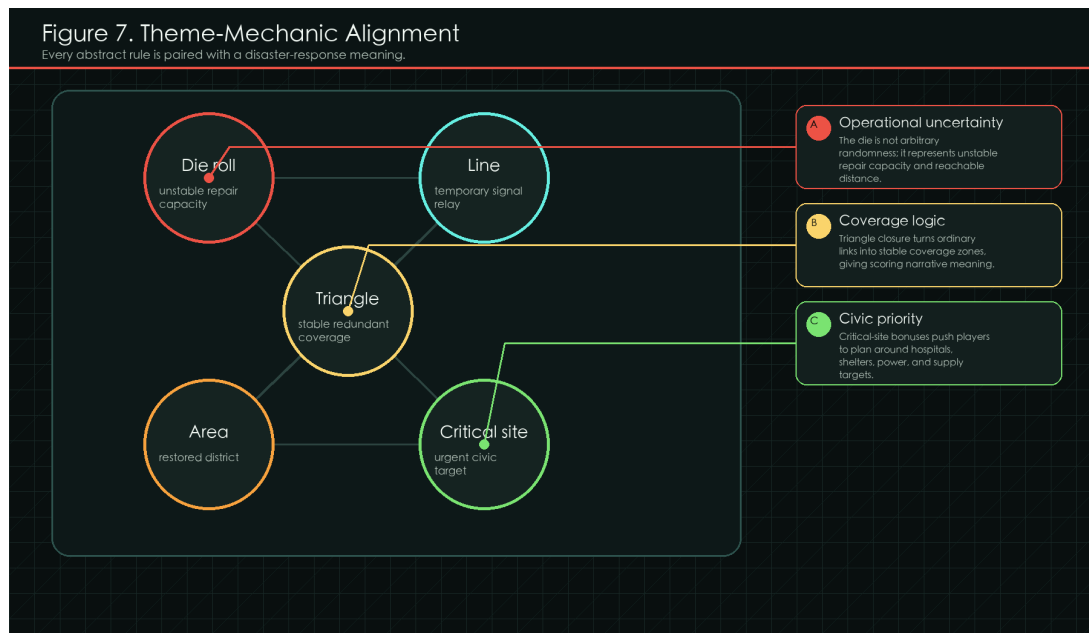


Figure 7. Theme-mechanic alignment: each abstract rule is paired with a disaster-response meaning.

Iteration and Reflection

The most important redesign choice was simplification. Earlier versions considered cards, roles, attacks, and special powers, but these made the prototype feel generic. I removed them so that meaningful decisions come from geometry: where to place a line, whether to close now or build larger, and whether a move gives an opponent an easy perimeter.

Playtesting and observation also changed the digital implementation. A balanced dice stream was added so one team would not repeatedly receive stronger rolls in a short match. Claim animations were cleaned up, end-game results were moved onto the map, and large-triangle recognition was expanded so the rules matched player expectation. Overall, Signal Aftershock makes a simple action - drawing a line - feel meaningful because every line changes the strategic shape of the board and the story of a city being reconnected.

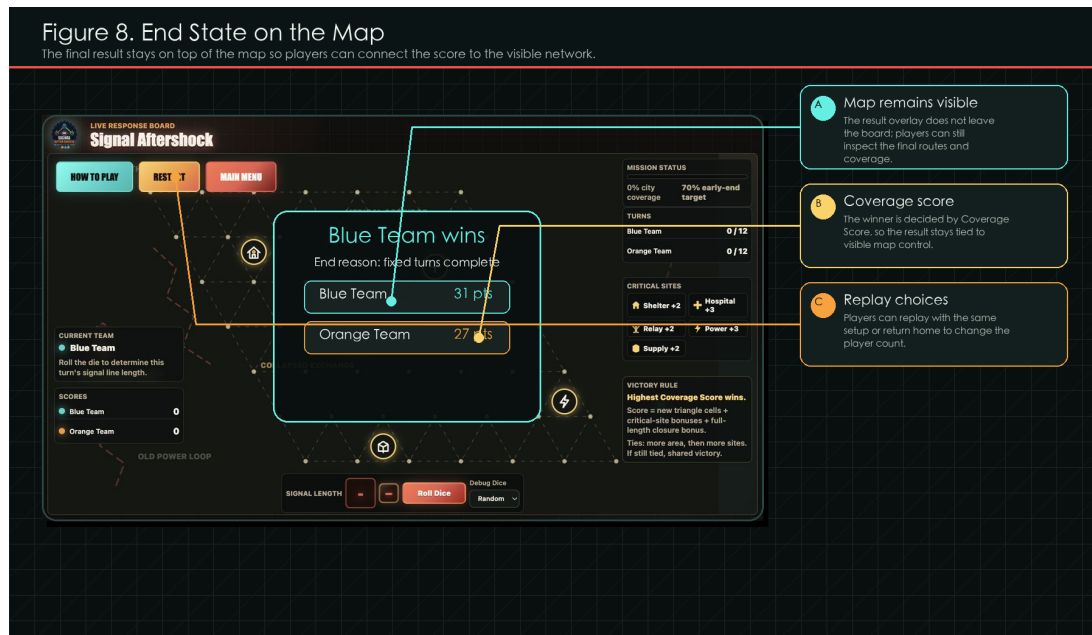


Figure 8. End state on the map: the final result stays on top of the board, allowing players to connect the score to the visible network.

AI Declaration

I used OpenAI Codex 5.3 coding assistance for the technical development of the playable browser prototype. My use of Codex focused on web-coding and implementation: writing and revising HTML, CSS, JavaScript, and SVG code for the interactive triangular grid, line drawing, hover previews, turn interface, scoring display, visual feedback, animations, and deployment-ready prototype files.

I worked through an iterative web-coding process. After testing the prototype in the browser, I identified layout, interaction, visual feedback, or bug issues, then used Codex to help generate code-level revisions. I reviewed the suggested changes, adapted them where necessary, updated the prototype, and tested the revised version again. This process was used across multiple versions until the interface, gameplay flow, and visual feedback matched my intended design.

The core rules and mechanics were my own design decisions. All Codex-assisted code and prototype files were reviewed, adapted, tested, and integrated by me before submission.

Bibliography

Barile, M. n.d. "Dots and Boxes." MathWorld - A Wolfram Resource.

<https://mathworld.wolfram.com/DotsandBoxes.html>

Salen, K. and Zimmerman, E. 2003. Rules of Play: Game Design Fundamentals. Cambridge, MA: MIT Press. <https://mitpress.mit.edu/9780262240451/rules-of-play/>